that uses the data break facility. These signals are received as the level input to three DCD gates on each module. Two of these gates serve as a complementary input to the MA for the Data Address signal. An inverter in the module precedes the DCD gate on the clear side of the MA flip-flop; so the single address signal input either sets or clears the flip-flop. The third DCD gate connects to the 1 side of the MB flip-flop to receive the Data Bit signal directly. Control circuits within the computer provide trigger pulses to all these gates. Input signals must be at ground level to designate a binary 1 or −3v to designate a binary 0. These signals must precede the trigger pulse by at least 400 nsec. Each Data Address input represents 12 ma of load. Each Data Bit input represents .11 ma of load.

R650 BUS DRIVER

The AC, MB, Break, and Run(1) output signals are buffered by bus driver circuits of Type R650 modules before connection to the interface.

The R650 contains two inverting bus drivers for driving heavy current loads to either ground or negative voltages. The bus drivers operate at frequencies up to 2 mc with typical rise and fall times of 25 nsec. The typical total transition times are 60 nsec for output rise and 65 nsec for output fall.

By grounding pin H or S the rise and fall time can be increased to avoid ringing on exceptionally long lines. The driver then operates at frequencies up to 500 kc with typical rise delay of 50 nsec, fall delay of 50 nsec, and total transition time of 800 nsec for output rise and 700 nsec for output fall. Terminal K or U can be used for driving coaxial cable.

The direct output (terminals J and T) drives 20 ma of external load at either ground or −3v. The resistor output (terminals K and U) drives 90-ohm coaxial cable such as RG-62-U. This output drives 5 ma of external load at either ground or −3v. The direct output connects to the interface connectors of the PDP-8.

S107 INVERTER

In the basic computer, Type S107 Inverter modules receive the Increment MB and Cycle Select signals used with the data break. In the Memory Extension Control Type 183 option the S107 Inverter receives the Address Extension 1-3 signals, and in the Data Line Interface option Type 681 it receives the Teletype Line(1) signal. Within these two options the Type S107 supplies the Data Field and Teletype Instruction (TT Inst) output signals.

The S107 Inverter contains seven inverter circuits with single-input diode gates. Six of the circuits are used for single-input inversion; the seventh circuit can be used for gating by tying additional diode input networks to its node terminal. Clamped load resistors of 5 ma are a permanent part of each inverter. Typical output total transition times are 60 nsec for rise and 50 nsec for fall.

The diode input accepts standard level inputs of ground and −3v that are a minimum of 100 nsec in duration. This 1 ma input load is shared among the inputs at ground. The node terminal input accepts only R001 or R002 Diode Network output connections, or their equivalents. The combined length of all leads attached to the node terminal must not exceed 6 inches. Input signal and load characteristics for diode networks are the same as those given for the diode input above.

Output signals from the inverters are standard levels of ground and −3v. Each inverter drives 15 ma of load at ground. Output terminals of inverters may be connected in parallel. Only one clamped load resistor is needed at the output when less than 2 feet of wire is used. If the wire exceeds this length, additional clamped load resistors may be necessary for a fast enough fall time in high frequency applications.

S111 DIODE GATE

Type S111 modules in the computer receive the Program Interrupt Request and Transfer Direction (Data In) signals.

The S111 Diode Gate contains three diode gates, each connected to a transistor inverter. The gate operates as a NAND for negative inputs and as a NOR for ground inputs. Each gate has three input terminals: two are connected to diodes; a third is connected directly to the node point of the diode gate. The third terminal allows the number of input diodes to be increased by adding external diode networks such as the R001 or R002 modules. External diodes must be connected in the same direction as the diodes in the S111. Typical output total transition times are 60 nsec for rise and 50 nsec for fall.

Input signals to the diode terminals must be standard levels of ground or −3v, and must have a minimum duration of 100 nsec. Input load is 1 ma shared among the inputs at ground. Input signals to the node terminals accept only connections from Type R001 or R002 Diode Network modules, or their equivalents. The maximum combined length of all leads attached to a node terminal is 6 inches. Input signal load is similar to the diode input.

S151 BINARY-TO-OCTAL DECODER

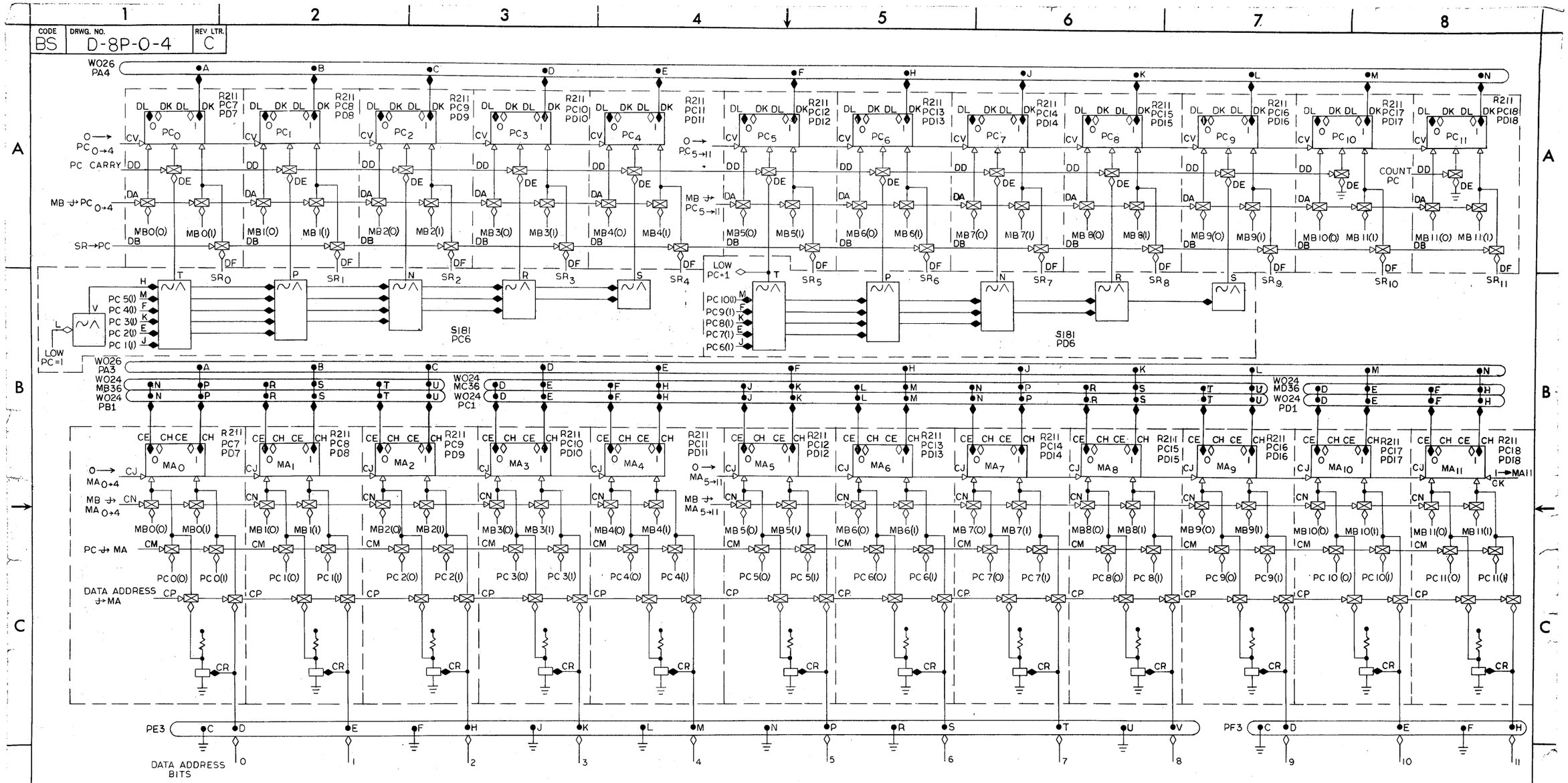
A Type S151 module (in parallel with a Type S107) receives Address Extension 1-3 signals supplied to the Memory Extension control option Type 183.

This S151 decodes binary information from three flip-flops into octal form. When the enable input is at ground, the selected output line is at ground and the other seven outputs are at −3v. When the enable input is at −3v, all outputs are at −3v. The internal gates are similar to those in the S111. The enable input is the common emitter connection of the output inverters. Typical total transition times are 75 nsec for output rise and 60 nsec for output fall.

Standard input levels are −3v and ground, 100 nsec minimum duration. The 2.3 ma input load is shared among the inputs at ground.

S203 TRIPLE FLIP-FLOP

The S203 contains three identical flip-flops. Each flip-flop has a direct clear and a DCD gate for conditional readin. The level input to one of these gates connects to the interface to receive the Data Break Request signal. This input receives standard ground and −3v levels. The conditioning level must be ground level to condition the gate and to request a break. This conditioning must occur at least 400 nsec before an internal computer pulse triggers the gate. The level input represents a 2-ma load at ground.



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B	Drawing no. BS-D-8P-0-3	6-8-65	
C	19	10-65	

REVISIONS

DRAWN	D. Wright	DATE	8-12-64
CHECKED		DATE	3/12/65
ENG		DATE	5/1/65
PROJ ENG		DATE	
PROD		DATE	

digital EQUIPMENT CORPORATION
MAYNARD, MASSACHUSETTS

ASSY NO

SHEET OF

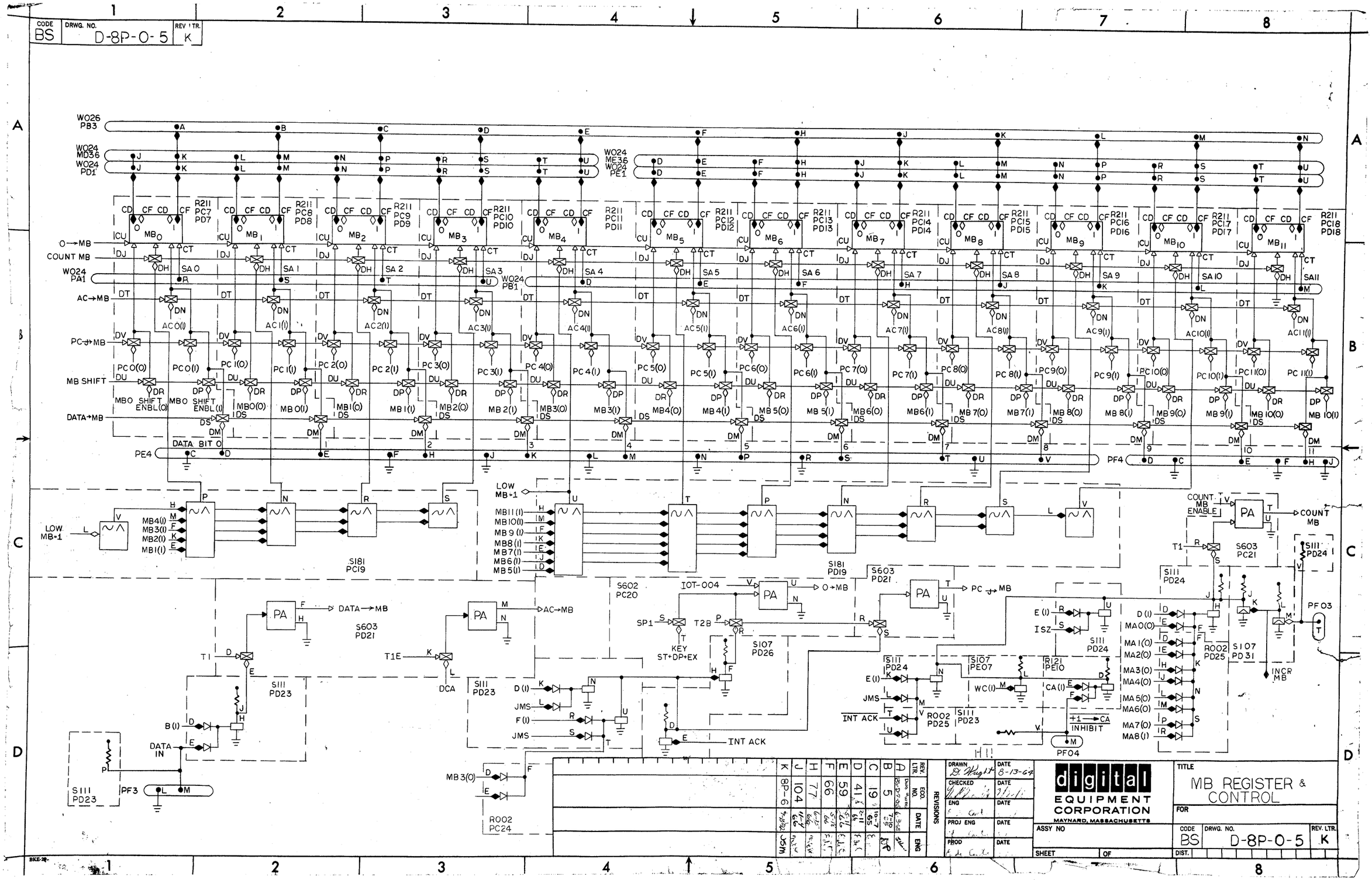
TITLE PC+MA REGISTERS

FOR

CODE	DRWG. NO.	REV. LTR.
BS	D-8P-0-4	C

DIST.

BKE-2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WO24 PB1
O→MB
COUNT MB
AC→MB
PC→MB
MB SHIFT
DATA→MB
DATA BIT O
PE4
PF4
R211 PC7 PD7
R211 PC8 PD8
R211 PC9 PD9
R211 PC10 PD10
R211 PC11 PD11
R211 PC12 PD12
R211 PC13 PD13
R211 PC14 PD14
R211 PC15 PD15
R211 PC16 PD16
R211 PC17 PD17
R211 PC18 PD18
LOW MB=1
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S181 PD19
S602 PC20
S603 PD21
S603 PC21
S111 PD24
S111 PD23
S107 PD26
S107 PD31
S107 PE07
R121 PE10
R002 PD25
R002 PC24
IOT-004
SP1
KEY ST+DP+EX
T2B
T1
T1E
DCA
DATA IN
PF3
PF03
PF04
INT ACK
INCR MB
INHIBIT
COUNT MB ENABLE
digital EQUIPMENT CORPORATION
MAYNARD, MASSACHUSETTS
TITLE MB REGISTER & CONTROL
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REV. LTR. | ECO. NO. | DATE | ENG
K 8P-6
J 104
H 77
F 66
E 59
D 41
C 19
B 5
A RS-P-2-35